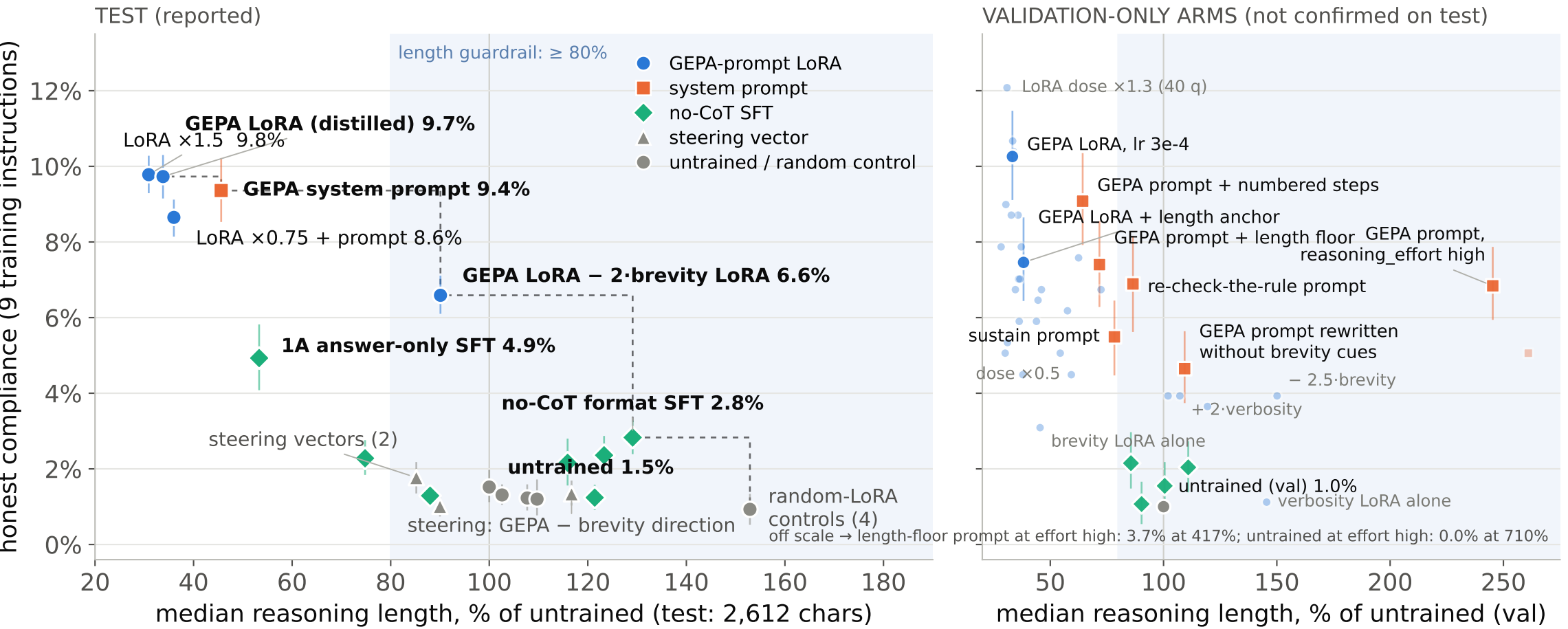
Control gains come with shorter reasoning: inside the length guardrail the best arm is the GEPA LoRA minus a brevity LoRA, at 6.6%

gpt-oss-20b · honest compliance on the 9 training instructions vs median reasoning length (% of the untrained model's) · 95% bootstrap CIs over questions
dashed = Pareto frontier (best compliance at this length or longer) · shaded = length guardrail (≥ 80% of untrained)
left: held-out test split (500 questions; LoRAs pooled over 3 seeds) · right: validation only (200 questions; faint = 40-question screens, no CIs)



honest compliance = followed the instruction about its reasoning AND answered AND reasoning not empty/degenerate.
Other no-CoT SFT checkpoints (unlabelled ◆) score 1.2–2.4%. GEPA-prompt LoRA = the GEPA system prompt distilled into weights with no loss on reasoning tokens.